

Student ID 1 & 2: (Do this in pairs)

Minimum Path Sum rules for this worksheet:

- You are given a grid of nonnegative costs.
- You start at the top-left cell and want to reach the bottom-right cell.
- You may only move one step right or one step down.
- The goal is to minimize the total sum of costs along the path, including both endpoints.

1. Warm-up: compute the path sum

For each grid, write the minimum path sum from the top-left corner to the bottom-right corner.

Grid	Minimum sum
1 2 3 4	
1 5 1 2 2 2	
2 1 3 4 0 2 1 1 1	

2. Choose a DP state

Let the grid have m rows and n columns, and define a table $\text{DP}[1..m][1..n]$. Complete the state definition:

$\text{DP}[i][j] =$ _____

3. Fill a Minimum Path Sum table

Use the grid from Homework 5:

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1 3 1 5 1
2 2 4 1 5
5 0 2 3 0
6 1 1 2 1

```

Fill the full DP table.

	1	2	3	4	5
1					
2					
3					
4					

4. Base cases

What are the base cases for the starting cell, the first row, and the first column?

5. Write the recurrence

Write the recurrence for an interior cell (i, j) where $i > 1$ and $j > 1$.

6. Complexity

What is the overall time and space complexity of the bottom-up DP solution?